A Cubic Norm-Tower Certificate for the 3^8 Multiplicative-Order Class in Decimal Mountain Concatenations (OEIS A077187)

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Abstract

For prime p with $\mathrm{ord}_p(10) = 3^8 2^a 5^c$, we establish existence of a mountain concatenation divisible by p, conditional only on explicit, reproducible exact-integer computations and the affine-monodromy lemma from the preceding project stages. A cubic relative-norm tower computes the 486 exact cyclotomic norms without large Sylvester resultants. A complete one-shot gcd sieve yields three spurious candidate primes, each eliminated by its true multiplicative-order phase. This is an infinite-subfamily result, not a proof of the full OEIS A077187 conjecture.

1 Mountain concatenations and affine drift

Define

$$M(k) = 1 \parallel 2 \parallel \cdots \parallel k \parallel (k-1) \parallel \cdots \parallel 1, \qquad r = \mathrm{ord}_p(10).$$

For $K_e = 10^e - 1$ let $C_e = 1 \parallel \cdots \parallel K_e$, $R_e = K_e \parallel \cdots \parallel 1$, $D_e = \sum_{j=1}^{K_e} \ell_{10}(j)$, $H_e = 10^{D_e}$ and $Y_e = H_e^{-1} R_e \pmod p$. Put

$$(U, V) = (C_{2r} - C_r,\ Y_{2r} - Y_r).$$

The previously established affine-monodromy identity [2] gives

$$(C_{jr}, Y_{jr}) = (C_r, Y_r) + (j-1)(U, V) \pmod p \quad (j \geq 1). \tag{1}$$

The normalized residues of $M(10^{jr})$ and $M(10^{jr}+1)$ equal respectively $10C_{jr} + Y_{jr} + 1$ and $1000C_{jr} + Y_{jr} + 121$. Since $p \nmid 990$ for the present order family, their slopes $10U + V$ and $1000U + V$ cannot both vanish unless $(U, V) = (0, 0)$. Thus nonzero drift produces a witness among $k = 10^{jr}$ or $10^{jr} + 1$, $1 \leq j \leq p$.

Set $r = 6561s$ where $s = 2^a 5^c$, $t = 729$, $m = r/t = 9s$ and $\gamma = s^{-1} \pmod t$. Let $\zeta = 10^m \pmod p$, of order t. The drift coordinates are quadratic in m, with coefficients in $\mathbb{Q}(\zeta)$. Their integer-scaled forms have denominator $D = 200t^2$. Set

$$E_0 = 0,\ E_j = \sum_{h=1}^{j} h 10^{h-1} \pmod t, \quad n_j = t/\gcd(j, t),$$

$$h_j = \frac{t}{n_j} \sum_{l=0}^{n_j - 1} l \zeta^{j(l+1)}, \quad j_j = \zeta^{-j} h_j^2, \quad \alpha_j = \zeta^{\gamma j 10^{j-1}}.$$

Write $DU_\gamma = U_0 + U_1 m + U_2 m^2$ and $DV_\gamma = V_0 + V_1 m + V_2 m^2$. The exact full-digit-layer recurrence yields $U_0 = 99t^2$, $V_0 = -81t^2$, and

$$U_1 = -90t^2 + \sum_{j=1}^{t-1} \zeta^{-\gamma E_j} [D(\alpha_j - 1) - 180 t \alpha_j h_j],$$

$$U_2 = 200 \sum_{j=1}^{t-1} \zeta^{-\gamma E_j} (\alpha_j - 1) j_j,$$

$$V_1 = 90t^2 + \sum_{j=1}^{t-1} \zeta^{\gamma E_{j-1}} [200 t \alpha_j h_j - 20 t h_j],$$

$$V_2 = 200 \sum_{j=1}^{t-1} \zeta^{\gamma E_{j-1}} (1 - \alpha_j) j_j.$$

The exact coefficient code computes these expressions in $\mathbb{Z}[\zeta_t]$; independent finite-field evaluation checks all primitive phases.

2 Algebraic obstruction and exact norm tower

Put

$$A = U_2V_0 - U_0V_2, \quad B = U_2V_1 - U_1V_2, \quad C = U_1V_0 - U_0V_1, \quad Q_\gamma = A^2 - BC.$$

Here Q_γ is the resultant in m of DU_γ and DV_γ. Simultaneous zero drift modulo p forces

$$p \mid N_\gamma = \left|\mathrm{Norm}_{\mathbb{Q}(\zeta_t)/\mathbb{Q}}\, Q_\gamma\right|_{\{2,3,5\}\text{-free}}.$$

The denominator D is invertible modulo all primes with order divisible by 6561, and p is unramified in the cyclotomic extension.

Lemma 1 (Cubic relative norm)**.** *If $x = \zeta_{3^n}$, $y = x^3$ and $W = A(y) + xB(y) + x^2C(y)$, then*

$$\mathrm{Norm}_{\mathbb{Q}(x)/\mathbb{Q}(y)}\, W = A^3 + yB^3 + y^2C^3 - 3yABC.$$

Proof. Multiply the three conjugates $A+xB+x^2C$, $A+\omega xB+\omega^2x^2C$, and $A+\omega^2xB+\omega x^2C$ where $\omega^3 = 1$ and $1+\omega+\omega^2 = 0$. □

Iterating the lemma computes the exact norm in degrees $486 \to 162 \to 54 \to 18 \to 6 \to 2 \to 1$; the final quadratic norm of $a + b\zeta_3$ is $a^2 - ab + b^2$. The complete certificate comprises 486 nonzero exact integer norms; all are computed using integer operations and checked against independent finite-field identities. Six full norms from the previous 3^7 stage match byte-for-byte.

Lemma 2 (One-shot smooth-order sieve)**.** *For $N > 1$ with $\gcd(N, 30) = 1$, write $W = \lfloor (N-1)/6561 \rfloor$ and, when $W \geq 1$, choose $A = \lfloor \log_2 W \rfloor$, $B = \lfloor \log_5 W \rfloor$ and $S = 2^A5^B$. Any prime $p \mid N$ with $\mathrm{ord}_p(10) = 6561 \cdot 2^a5^c$ divides*

$$\gcd\left(N,\ x^2 + x + 1\right), \qquad x = 10^{2187S} \pmod N.$$

Proof. As $6561 \cdot 2^a5^c \mid p - 1 < N$, both 2^a and 5^c are bounded by W and divide S. Consequently x has exact order three modulo p and satisfies $x^2 + x + 1 \equiv 0 \pmod p$. □

Theorem 1 (The complete 3^8 smooth-order subfamily)**.** *For every prime p with $\mathrm{ord}_p(10) = 6561 \cdot 2^a5^c$ $(a, c \geq 0)$, there is a positive integer k such that $p \mid M(k)$. Moreover k can be chosen as 10^{jr} or $10^{jr} + 1$ with $1 \leq j \leq p$.*

Proof. For every one of the 486 primitive phases γ modulo 729, the exact norm N_γ computed using Lemma 1 is positive. Applying Lemma 2 to each norm, the 486-entry GMP certificate has exactly six nontrivial rows (all others have gcd 1):

γ	gcd
202	328051
247, 266, 346, 458	52489
698	$328051 \cdot 839809$

The prime factors have the following exact multiplicative orders and true phases:

p	$\mathrm{ord}_p(10)$	$s = \mathrm{ord}_p(10)/6561$	$s^{-1} \bmod 729$
52489	26244	4	547
328051	328050	50	452
839809	52488	8	638

In all three *true* phases 547, 452 and 638, the corresponding gcd is 1. Thus none is a genuine simultaneous zero-drift case. As every potential prime must divide one of these gcds, $(U, V) \neq (0, 0)$ for the target order class. Equation (1) then yields the asserted witness in at most p phases. □

3 Scope and reproducibility

The accompanying program regenerates exact cyclotomic coefficients and all 486 norms, evaluates the 486 gcds using C++/GMP, and checks the apparent factors against their exact orders and phases. A second recurrence verifies three representative drift vectors. This is a computer-assisted *partial* theorem: no proof is asserted for arbitrary p, especially when $\mathrm{ord}_p(10)$ contains prime factors outside $\{2, 3, 5\}$ or $v_3(\mathrm{ord}_p(10)) \geq 9$. No external referee review or OEIS edit has occurred.

References

[1] OEIS Foundation, *A077187*, https://oeis.org/A077187 (accessed October 17, 2026).

[2] Q. Wang, *Eliminating the 3^7-order exceptional set for decimal mountain concatenations*, exact norm and affine-monodromy certificates, 2026.